

READYKIT · PROBLEM STATEMENT

ReadyKit problem statement

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/readykit		
Source	docs/01-problem.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-09-26	Amish Chadha	D5 reversed by Amish (RDk-DDR-002); license back to CERN-OHL-S-2.0; constraint line describes the portfolio license pair
0.3	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.2	2026-09-25	Amish Chadha	Users, context, prior work with sources, constraints and out of scope for TRL 2
0.1	2026-09-25	Amish Chadha	Initial scaffold

ReadyKit problem statement

Small hardware teams and open projects rarely document requirements, decisions and readiness consistently, so work is hard to review, reproduce or fund. The tools that exist cover pieces of the job (assembly instructions, metadata, certification) but no free tool ties document control, readiness levels, drawings and concept renders together in one repository with a check that runs on every push.

■ The problem in numbers

- In a study of 132 open source hardware products, only 11 (8%) published all eight documentation elements the authors assessed, such as editable CAD, a bill of materials and assembly instructions; the mean score was 4.2 of 8 ([Bonvoisin et al. 2017](#), *Journal of Open Hardware*).
- In a *Nature* survey of about 1,500 researchers, more than 70% said they had tried and failed to reproduce another scientist's experiment ([Baker 2016](#), *Nature* 533).
- Funders ask for readiness in a common language. NASA defines Technology Readiness Levels 1 to 9 ([NASA](#)), and Horizon Europe uses the same scale in its calls ([Horizon Europe NCP portal](#), [TRL self-assessment guide](#)). Small teams seldom keep the evidence that would support a claimed level.

■ Users and context

User	Situation	What they need from ReadyKit
Solo maker or two-person team	Builds in a garage or makerspace; documents in a README, if at all	A repo that starts documented, and a check that says what is missing
University lab or student team	Needs reviewable designs for supervisors, theses and competitions	Controlled documents with versions and revision history that print cleanly
Open science hardware group	Shares instruments with other labs (GOSH, AfricaOSH, reGOSH networks)	Documentation others can rebuild from, with a stated readiness level
Early-stage hardware startup	Prepares grant or accelerator applications that ask for a TRL	TRL claims tied to evidence files, checked in CI
Team using AI coding agents	Lets an agent draft documents and CAD	Written guardrails the agent reads, plus automatic checks that catch drift

The kit was built for the Design Molecule open hardware portfolio of about 85 project codes and is in daily use there. ReadyKit is the proposal to release it as a standalone, reusable tool.

■ Prior work

Tool or standard	What it covers	Gap ReadyKit addresses
GitBuilding	Markdown-based assembly instructions for open hardware	No document control, TRL gating or CAD-driven media
DIN SPEC 3105-1	Requirements for open hardware technical documentation	A specification, not a tool; no automatic check
Open Know-How (OKH)	Metadata manifest so designs can be found and indexed	Describes a project; does not produce its documents
OSHW certification	Self-certified openness of a hardware project	Licensing and openness, not document quality or readiness
build123d	Python CAD library	Geometry only; ReadyKit builds sheets and renders on top of it

■ Constraints

- Hardware budget \$0: ReadyKit is software, free to use and free of paid services.
- Runs on a normal laptop with Python 3.11 or later; no GPU and no CAD license needed.
- Plain text in Git as the source of truth (Markdown, YAML, Python), so every change is reviewable in a diff.
- Works offline, except for installing packages and the 3D viewer page, which loads its viewer script from a public CDN.
- Open licenses throughout: the portfolio pair used on every repo, CERN-OHL-S v2 for hardware and non-code content (documents, drawings, CAD, BOM and media) and MIT for software (RDK-DDR-002); fonts SIL OFL.

■ Out of scope at this stage

- A hosted web service or account system.
- Formal quality management certification (for example ISO 9001) or regulatory submissions.
- Electronics design checks (KiCad) and firmware documentation beyond linking to the files.
- Any claim that the kit makes a design safe or fit for use; it checks documentation, not engineering.